

Situated Buttons: A User Interface to Support Users with Complex Needs and Promote Independent Living

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ABSTRACT

We present *situated buttons*, a highly personalized, affordable system to support users with complex needs such as people with learning disabilities and autism, dementia, etc., and promote independent living. The system addresses concerns many users with mental health issues have with accessing assistive services and technologies as a result of their health condition. The solution consists of a wireless push button that via a simple click can launch a video on a device such as a smartphone or a tablet. A trusted person like a family member or carer can appear in the video to illustrate how to complete certain activities of interest to the end-user. The button can be placed anywhere where support is needed, e.g. on a washing machine or at the kitchen table. We conducted workshops and online discussions with social and healthcare organizations, potential end-users (people with learning disabilities and older adults), and their carers. The results show that our solution has the potential to support individuals with complex needs, improve independence, and increase their quality of life, by giving simple access to helpful instructions in their daily life.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices.

KEYWORDS

Smart buttons, Assistive Technology, Disability, Personalization

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1 INTRODUCTION

Assistive technologies can support and empower users suffering from a wide range of health conditions [39–41]. Every-day and off-the-shelf technologies have the potential to support people with

complex needs, such as people suffering from learning disabilities, autism, and dementia. Smart home appliances such as smart light bulbs, personal assistants like Amazon Alexa, smartphones, etc., are readily available and affordable and can support people with complex needs in their daily life. However, they can encounter difficulties in interacting with such devices due to their condition or the lack of confidence in their ability to properly use devices. As a concrete example, the recollection of keywords and the short amount of time in which users need to verbalize the personal assistant's commands (e.g., “Alexa/Hey Google, turn on the living room lights.”) can become problematic for users with learning disabilities, speech impediments, or dementia [3, 4].

We present the concept of *situated buttons* - smart buttons located in *specific places* and enabling *highly personalized videos* to help people with complex needs maintain and develop new skills, in the context of their home (Figure 1). Previous studies focused on situated tools such as special boxes located in community day centers [49] or for use in therapeutic settings [50] for adults with cognitive disabilities, or containing smart buttons linked to Android applications for children with learning disabilities in the context of their home. Our system differentiates from previous studies in 2 ways. Firstly, the wireless smart button is located in a specific place where it is needed (e.g., on a washing machine for doing laundry), also acting as a reminder of its purpose. Secondly, via a simple push, the system can provide highly personalized content such as videos with family members, friends, carers, or even the end-users illustrating how to complete activities referring to personal care (e.g. nutritional support, planning to dress appropriately for weather), engaging in interests and activities, and maintaining environmental hygiene (e.g., doing laundry). Whilst the content launched can also be generic YouTube videos or applications illustrating how to conduct tasks, we argue that this solution introduces a higher level of personalization since all instructions are explained by a trusted person, are tailored to an end-user's requirements, and in the context of their home. To validate the concept, we conducted workshops and online discussions with potential end-users, city council officers, social care, and healthcare professionals.

This paper is structured as follows. We start by providing an overview of smart buttons and assistive technologies research studies, with a focus on the healthcare domain. We then present the evaluations we conducted with social and health care organizations, people with learning disabilities and autism (LDA), and frail and older people. Finally, we discuss the general implications and the potential for future work.

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Figure 1: Examples of situated buttons with visual reminders of their functionality. A video with indications of how to prepare a sandwich (left) and use the washing machine (right) will appear on the screen when the button is pressed.

2 BACKGROUND

In this section, we outline the current research in the area of smart buttons and assistive technologies. We focus on how these technologies can support independent living and improve the quality of life of people living with various conditions and the issues they might face when interacting with such devices.

Tangible/physical buttons are gaining popularity and are widely used as an input device in the Human-Computer Interaction field [5]. Buttons can serve multiple purposes. For example, panic or emergency buttons can be used as pendants [24] or incorporated in armbands to facilitate independence for people with dementia [6]. Research shows that buttons are useful in supporting people with visual impairments during everyday tasks [25] or to improve women's safety [7]. Studies have also been conducted regarding the location of the button and the concept of wearable buttons, such as buttons embedded into clothes [8, 22]. Cheung and Girouard [8] show that importance should be given to the location of the wearable button as some people would prefer having a health-related button close to the heart and a smartwatch-like button close to the wrist.

Previous research studies illustrate how technology using visual models and auditory prompts can assist people with complex needs in living independently and increase their quality of life [13-16]. Bassette et al. [13] show that mobile phone applications containing videos illustrating physical exercises are suitable for adolescents with Autism Spectrum Conditions. The application in association with human prompting and reinforcement can motivate adolescents to be physically active. Audio and picture prompting has been successfully used to enhance daily living skills (e.g., washing dishes and using the washing machine) of adults with Autism Spectrum Conditions [15]. The human voice could also be useful for people with verbal and memory problems [19] and can be beneficial in learning interventions. Milani et al. [20] show that audiobooks can reduce the emotional-behavioral problems of adolescents with dyslexia and can have a beneficial effect on reading accuracy. Yonan and Sommers [21] also indicate that older adults can benefit from

listening to familiar voices as it can help them identify words in sentences.

Assistive technology interventions can be effective; however, they need to be flexible and customized to the individual [12, 16, 23, 45]. Dawe [23] highlights that simplicity, in terms of functions, installation, and maintenance is crucial for young adults with cognitive disabilities. The author gives the example of a non-verbal young adult with autism using 8 buttons, each button playing a voice recording such as *hello* or *goodbye*. Intuitively, having multiple buttons becomes challenging to use. Similarly, Vermeer et al. [24] indicate that people with dementia do not want too many buttons due to the learning required and the risk of activating the button by mistake, especially in the case of panic buttons.

3 SITUATED BUTTONS

Situated buttons are stand-alone physical buttons that can be used to proactively trigger events, thus providing a shortcut to a command and avoiding the use of the complex voice assistant syntax or smartphone applications. For our prototype, we used Flic [2], a wireless push button that proved successful when undertaking studies with children suffering from profound learning difficulties [48]. The button can control applications, devices, and services (e.g., YouTube) and carries out 3 functionalities via clicking, double-clicking, or holding on the button. We customized the button so that via a simple click, a personalized video can be launched on a smart device such as a smartphone or a tablet (Figure 1). A family member, a friend, a carer, or even the end-user can be part of the video and illustrate how to complete activities, such as preparing a sandwich (Figure 1 left) or using the washing machine (Figure 1 right). This solution of using buttons and videos for prompting and supporting individuals to enhance their skills introduces a higher level of personalization than a generic YouTube video as all the instructions provided are tailored to an end-user's requirements and in the context of their home. The functionality is agile; once a skill is developed, it can easily be changed by simply creating a new video of a new desired skill. Every person has different needs, lifestyles, and health conditions that change over time [45].

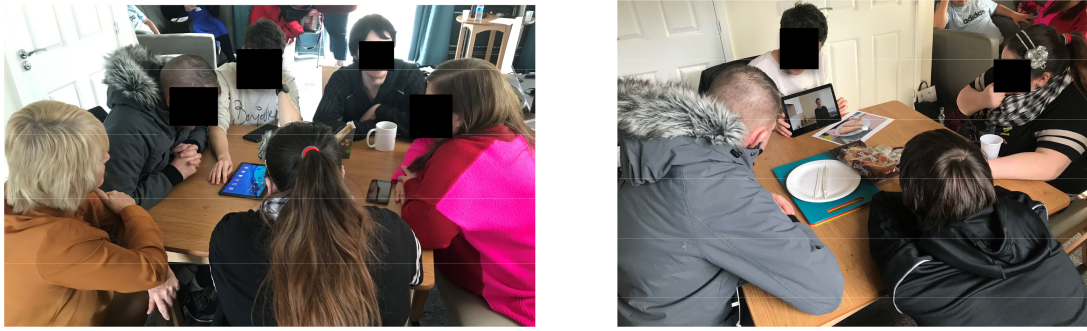


Figure 2: Participants taking part in the workshop – before (left) and during (right) the sandwich preparation.

Therefore, it is important to personalize the intervention as there cannot be a generic approach to social care [17, 18, 45]. With this in mind, the content that is triggered should be highly personalized for the individual's home, for instance by using a familiar voice and figure.

To evaluate the concept, we created a test video that would suit a wide range of users. We started by filming a video supporting a nutrition scenario (making a cheese sandwich) as this is an important area where people with learning disabilities and autism (LDA) and dementia require support [42]. Following the indications by Carnaby and Cambridge [1], the video was adapted to people with LDA and illustrates all the steps needed to make a sandwich, from collecting the equipment needed to washing hands, spreading the bread with butter, etc. Using the Flic application installed on an Android OS tablet, we configured the application so that when a user clicks on the button, the video with instructions on how to prepare the sandwich would appear. We intentionally placed the button on a sandwich image (Figure 1 left) to remind users about the purpose of the video.

3.1 First Validation: End-user Workshop

To validate our concept, we conducted a workshop with 4 potential end-users. These were people with mild LDA (3 males and 1 female), accompanied by 4 carers. A workshop was selected as the User-Centred Design methodology proved successful when designing and evaluating applications for mental illnesses [43]. The workshop took place in the same test house where the video was filmed. The end-users were customers of the housing association that owned the house and were living in a shared home. Meetings were held prior to the workshop with one of the carers to ensure that our workshop structure is appropriate and does not cause any emotional distress to the participants. Additionally, together with the carer, we carefully designed an easy read version of the information sheet handed in prior to the workshop.

The workshop lasted for 2 hours and was divided into 2 phases: (1) *Phase 1* – end-users and their carers started by familiarizing themselves with the button, followed by an open discussion about the possibility of using it in their daily life; (2) *Phase 2* – end-users were asked to view the video illustrating how to prepare a sandwich, to follow the instructions, and prepare the sandwiches on their own (Figure 2). This was necessary to understand end-users' views regarding the concept and whether they are able to follow

the instructions provided in the personalized video. Participants were provided with similar equipment (e.g., plates) and ingredients used in the video. Notes were taken down and data collected were analyzed using elements of thematic analysis [35].

3.1.1 Findings. Preliminary results show that the concept of a smart button enabling highly personalized videos is desired by end-users and their carers. Several themes emerged from the workshop held, as follows:

Independence. None of the end-users and their carers have previously used a button to support their daily needs. All participants provided positive feedback and mentioned that the videos could be used to enhance a variety of skills such as washing up or brushing teeth. One participant not knowing how to operate the touch controls on a hob (leading to frustration as they press it many times without it turning on), mentioned:

I think it's magic. I could use it to show me how to use the hobs. I can't do the hobs and I don't feel independent.

Monitoring. The end-users also indicated that the button would be useful to notify staff members to request assistance. Research studies [26, 27, 45] investigated the use of older people monitoring systems and the benefits they might bring in facilitating and supporting the network of people caring for the older person. Similarly, the support workers would like to be informed if the customers conducted their duties:

Wouldn't mind a button that shows she took her meds.

Information Sharing. End-users, on the other hand, might not want to share everything with their carers. HCI researchers have previously studied the willingness and the type of information people with complex needs would share with their carers [28]. Prasad et al. [29] indeed show that older people are less willing to share diet and exercise information with their carers than medication tracking. Similarly, one participant, who tends to overeat when they are angry and depressed although they know it is not good for their health, mentioned:

If I put a button on my fridge the staff would know, and they could come and stop me.

Ease of Use. Although they had not used a button before, most end-users owned a mobile phone and a smartphone and were familiar with the concept of watching videos on a screen. Participants

needed to pause the video multiple times to follow the instructions. Since the button only starts the video from scratch, participants had to pause and restart the video by tapping on the tablet. This can indicate that a single-button interface with Start / Stop / Restart Video options would be more appropriate for this type of interaction.

Collaboration. End-users were observed preparing the sandwich and no emotional distress was noticed. Each end-user was asked to prepare their sandwich. We wanted to create a pleasant environment and therefore, end-users prepared their sandwich together and at the same time, resulting in a positive and enjoyable experience (some end-users ate the sandwich at the end). We believe that this introduces the notion of collaboration and collaborative design which proved successful in maximizing the effectiveness of mental healthcare interventions [36]. Geeng and Roesner [37] aimed in understanding how multiple people living together interact with devices in smart homes. Cooperation but also tension arises within householders when installing, ordinary using the smart device or when the smart devices do not act as expected. Moreover, Trajkova and Martin-Hammond [39] show that older adults living in a community center encounter difficulties in interacting with the Amazon Echo, and most become non-users. One common reason for becoming a non-user is the need to negotiate use in a shared space. This can indicate that there is a need to differentiate between private and shared buttons. It could be that end-users might wish to have a private button for tracking their data (e.g., calory intake that they don't feel comfortable sharing with carers or even maybe with other residents) or for visualizing videos for developing personal hygiene skills. Shared buttons, on the other hand, located in shared spaces such as a kitchen can refer to common videos for all residents. Multiple buttons can also be placed; however, as suggested by [23, 24], simplicity is key, and designers should consider not overwhelming users with too many buttons or cause annoyance in other residents [45].

3.2 Second Validation: Online Workshop

The findings from the previous workshop indicated that the concept has great potential in supporting people with complex needs such as LDA. Therefore, we wished to determine whether it can also improve the quality of life of older adults and we filmed a second video in the same test house. We selected a scenario regarding hygiene – doing laundry – as this is an area where older and frail people require support [42]. As suggested in other research studies [8], we situated the button on the washing machine (Figure 1 right) to give users access to the information in the location where it would be needed. The instructions presented in the video were carefully selected together with a collaborator in this work, expert in the needs of older people.

Due to COVID-19 restrictions, it was not possible to hold face-to-face demonstrations for the second workshop. Similar to other research studies [10], we continued to seek feedback via internet multimedia platforms and conducted online discussions with 2 end-users - a frail old man living with his daughter and son-in-law who act as carers, and an older woman together with her live-in daughter, also acting as a carer. We also held discussions with 22 specialists from various councils, social care, and health organizations that

work with and support older and frail people, and people with mental health issues.

Similar to [11], both stakeholder categories were presented with the 2 videos illustrating the smart button functionality (making a sandwich and doing laundry) and were asked to provide feedback. The aim was to understand whether the concept of the smart button enabling a video can also empower older people to maintain independence and enhance their quality of life. The discussions were based on open-ended questions; notes were taken down and data collected were analyzed using elements of thematic analysis [35].

3.2.1 Findings. Positive reactions were noticed from both stakeholder categories. Various themes emerged, as follows:

Skills Enhancement. The specialists mentioned that the idea of using buttons and videos for helping people enhance their skills (e.g., cooking, washing clothes, using the microwave) has great potential:

I can see a benefit in watching themselves doing activities and using this as a reminder.

Independence. Similar to the previous workshop findings, despite not owning or previously interacting with a smart button, both end-users and their family members, believed that the system can provide independence and enhance the quality of life, including the life of the family members. The older man thought the idea of watching a video of his daughter using his microwave and having a video in the bathroom showing how to wash hands would be useful to him. Having a family member in the video would be ideal for him since he would not be able to make the video, given his frailty. The older woman also mentioned that she would like her daughter to appear in the video, but she also found it acceptable to download a video from the internet and having an unknown person in the video.

Ease of Use. With regards to other smart home devices, the older woman and her daughter were using a voice-enabled personal assistant such as Amazon Alexa. Interestingly enough, regarding cooking instructions, the older woman mentioned she would prefer the button and video to a voice-enabled personal assistant as she believes Alexa offers too much information that is '*difficult to retain in one go*'. The older man additionally mentioned he might need a prompt to use the button for starting the video.

Information Sharing and Monitoring. Carers would like to view remotely when and how many times the button is used by a customer. However, similar to the findings from the LDA workshop, there needs to be a balance between access to data and customer's right to privacy.

Configuration. Similar to Dawe [23], the specialists were particularly interested in the ease of adoption since off-the-shelf technologies such as smart buttons are widely available and can be easy to install:

Some of the current systems that are in use across services require considerable setup and procurement can be difficult. It's amazing that some of the off-the-shelf systems can be set up so easily to provide remote monitoring and support.

Personalization. As previously highlighted [6, 12, 45], the specialists explained that personalization is very important and it is vital to assess the needs of the customer. Furthermore, they believe that the smart button and enhanced video can ensure that customer support would be individualized. One specialist mentioned that it is important to assess the needs of the customer because individuals learn in different ways, and technology should reflect their preferred style of learning:

Some people are visual learners whereas others learn best through listening. It is good to have different options for carers to use with their client.

4 DISCUSSION

We argue that the *situated buttons* concept – where a smart button located in a *specific context* activates a *highly personalized instructional video* – has the potential to support skills maintenance or enable the end-user to develop new skills. Similar to [8], our results indicate that the location of the button is important. Buttons located in shared spaces can induce collaboration. Therefore, researchers should consider designing for shared spaces [46] and experiences that lead to joy and reduced conflict, e.g., residents wanting to use the same button at the same time can lead to annoyance and conflict [45].

Researchers might also wish to consider the purpose of the video and consider distinguishing between personal (e.g., referring to personal hygiene) and shared buttons. For example, it can be that end-users might wish to visualize videos referring to taboo subjects such as menstruation or different body parts [44] only in the comfort of their private room. Importance should also be given to data sharing (e.g., in cases where the end-user does not want to share data with their carers) so that the button is not seen as a conflicting tool between the needs of direct (end-users) and indirect (carers) stakeholders [47].

Results show that the button can also be seen as a tool in sharing information and even emotions (e.g., carers can know when a user is overeating and feeling depressed). Sharing emotions has been of great interest to the HCI community and research has been conducted on how to design for emotions [33], emotion awareness, and emotion regulation [28]. In the healthcare field, Jacob et al. [30] show that people suffering from cancer would not share emotional factors, such as loneliness with their oncologist. It is well known that feelings of anger and depression can have an impact on a person's quality of life and wellbeing, with previous research indicating that the ability to control one's emotions can improve mental wellbeing [31]. However, to regulate emotion, it is important for the person to be aware of the emotion [31] and to share it [32]. This can indicate that the button might have a secondary purpose, such as emotion sharing, which in time can help to regulate the emotion, improve mental wellbeing, and even strengthen the relationship with the carer.

Situated buttons is a highly personalized concept that enhances the functionality of a smart button by providing tailored functionalities based on the end-user's needs. Personalization is a key aspect of assistive technologies, especially in situations where a person's cognitive abilities are constantly changing [45]. Therefore, assistive technologies need to be agile and align with the new

needs of the end-user. Family members and carers can constantly change the purpose of the button to support skills development and maintenance. For example, end-users might wish to develop cooking skills one week and once they have mastered the skill, they might wish to learn how to do their laundry. The agility of the button can be embedded in an individual's habitual practices and support responsiveness to ever-changing needs and health conditions.

5 FUTURE WORK

We believe our solution has the potential to support individuals with complex needs and we wish to conduct further studies to evaluate our concept and its applicability to other cognitive conditions. As previously suggested, individuals have different requirements and learn in different ways. We wish to assess learnability [13, 15] and understand whether other personalized forms of presenting instructions implementable on a screen (e.g., images, images and text, images and verbalized text, or a combination of these) are also suitable. By this, we also aim to reduce the effort and the resources required to film the videos. Besides the visual reminders we already implemented, it could be that the end-users can also receive audio prompts for using and starting the button. Moreover, by conducting field studies in care homes, we wish to investigate whether the button can be viewed as a tool for sharing emotions and how to regulate possible arising tensions between monitoring and privacy.

6 CONCLUSION

We presented *situated buttons*, a highly personalized concept consisting of an affordable smart button linked to a tablet or a smartphone. Smart buttons can provide a wide range of functionalities that can enable people with complex needs to live independently. We believe that the button functionalities can be enhanced to transform it into a suitable technology to support individuals with complex needs to maintain and develop new skills. Via a simple click of the button, a video can be launched on a smart device such as a tablet or smartphone. A family member, a carer, or even the end-user can be part of the video and illustrate how to complete certain activities that are of interest to the end-user. This solution of using buttons and videos for prompting and supporting individuals to maintain and enhance their skills introduces a high level of personalization as all the instructions provided can be tailored to an individual's needs, lifestyle, and health condition.

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